

Classical Still Wins the Wall-Clock: A Quantum-vs-Classical Benchmark of Twenty Canonical Problems on Five Real Frameworks

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Abstract. Demonstrations of quantum algorithms often stop at a toy circuit and report no classical baseline. This report formulates twenty canonical quantum problems, attacks each with the real, dedicated frameworks (Qiskit with Aer, PennyLane, Cirq, Stim, and NumPy/scikit-learn for the classical baselines), and puts every quantum method next to its classical baseline with both costs on the table. The result, from 119 archived, reproducible traces, is unambiguous: at lab scale the classical baseline is faster in wall-clock time on *all twenty* problems, usually by two to four orders of magnitude, and this holds even for the flagship “advantage” algorithms, Grover’s search and Shor’s factoring, because their asymptotic advantages do not materialise at the small instance sizes a laptop simulator reaches. Yet the quantum phenomena are real, and the benchmark reproduces them: the CHSH value reaches the Tsirelson bound $2\sqrt{2}$ and violates the classical local-hidden-variable bound of 2 (but only with entanglement; a separable state gives 1.41), Grover uses provably fewer oracle queries (2 versus 7 on one instance), and Shor’s order-finding factors 15 correctly. In summary, on these twenty problems the quantum methods win at *physics* (nonlocality, query complexity) and not at *wall-clock computation* at the instance sizes a simulator reaches, and the benchmark shows why, with measured numbers on real frameworks, reproducibly.

Index Terms. Quantum computing, quantum advantage, benchmarking, Grover, Shor, CHSH, Bell inequality, QAOA, VQE, NISQ, reproducibility, classical baselines.

I. INTRODUCTION

DEMONSTRATIONS of quantum algorithms often omit the comparison that decides whether a method is useful: a tutorial runs a Bell pair or a two-qubit Grover and stops, and a benchmark reports a quantum method’s accuracy without running the classical baseline next to it. The research literature states that near-term quantum hardware is noisy and limited [1], [2], and the headline results (quantum supremacy on a sampling task engineered to be hard for classical machines [3]) concern a task chosen for classical hardness rather than a useful problem. This report makes the comparison concrete and reproducible: it takes the problems quantum computing is

usually taught on, runs them on the real frameworks, puts the classical baseline beside each one, and reports the measured numbers.

The laboratory behind this report, QLab, is a small Problem $\times$ Solver engine. A *problem* is a formulation (search, factoring, ground-state energy, a Bell test) with its instances and observable; a *solver* is a thin adapter over a real framework: Qiskit with the Aer simulator, PennyLane, Cirq, and Stim for the quantum methods, and NumPy with scikit-learn for the classical baselines. Every run records its cost (qubits, gates, shots, wall-time) and its result, and the traces are archived in the repository so that every number can be checked. Over twenty problems and 119 archived traces the engine answers one question for each: run on the best real framework available, does the quantum method beat the classical baseline, and at what cost? The answer is the subject of this report.

II. THE BENCHMARK: TWENTY PROBLEMS, FIVE FRAMEWORKS

The twenty problems span the standard curriculum: single-qubit and interference fundamentals; entanglement (Bell/CHSH, teleportation, superdense coding, state preparation); the oracle algorithms (Deutsch–Jozsa, Bernstein–Vazirani, Simon); the flagship algorithms (Grover’s search [4], Shor’s factoring [5], the quantum Fourier transform, quantum phase estimation); the variational methods (QAOA for MaxCut [7], VQE [8], a small quantum-machine-learning classifier); and noise and quantum error correction (a repetition code and a surface-code round, simulated with the dedicated stabilizer simulator Stim [12]). Each problem is solved by every applicable framework and by a classical baseline, under a fixed seed and shot budget, and the resulting trace records the outcome and the full cost. The costs are what make the comparison fair: a quantum method that gets the right answer is not “winning” if it needs a hundred times the wall-time of a classical routine that also gets the right answer, and the trace shows both side by side. All traces run on classical simulators of the quantum circuits, the regime a public laboratory can reach; this is the correct regime for the lab-scale question the report asks.

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Interactive laboratory and reproducible traces (MIT) at https://github.com/fsantibanezleal/CAOS_QLAB; the laboratory is available at <https://qlab.fasl-work.com>.

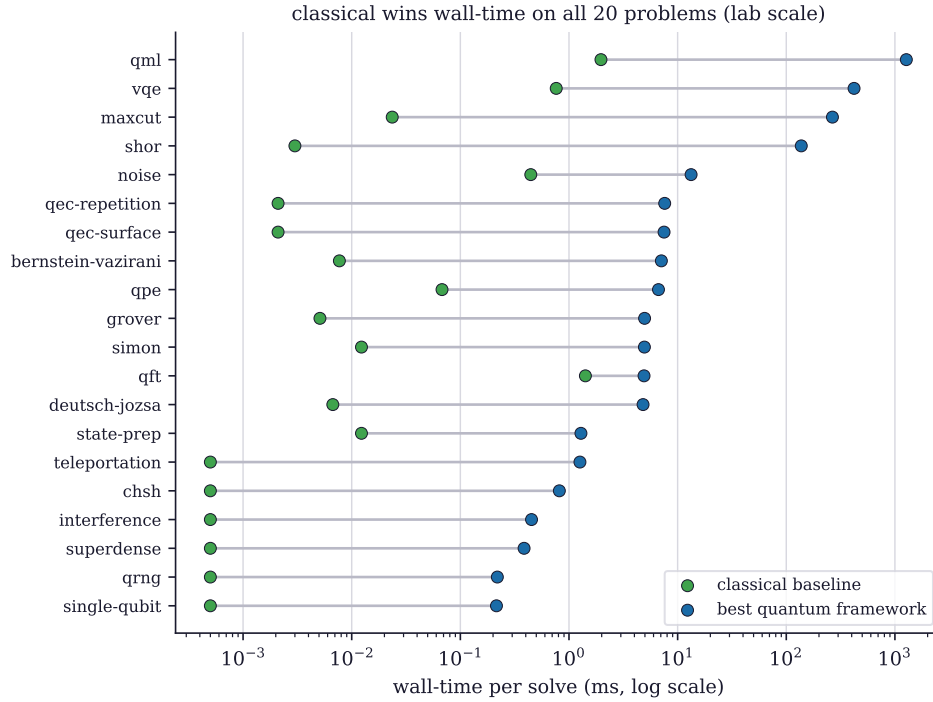


Fig. 1. Wall-time per solve, classical baseline versus the best quantum framework, on each of the twenty problems (log scale; problems sorted by quantum wall-time). The classical baseline (green) is faster on *every* problem, usually by two to four orders of magnitude, from microseconds against 0.4 to 7 ms on the oracle and entanglement cases to at most 2 ms against 266 to 1274 ms on the variational cases (QAOA, VQE, quantum machine learning). This includes the flagship algorithms: Shor factoring 15 and Grover search are both far cheaper to run classically at this scale.

III. THE WALL-CLOCK RESULT: CLASSICAL WINS ALL TWENTY

The headline result is Fig. 1 and Table I: on all twenty problems, the classical baseline is faster in wall-clock time than the best real quantum framework, at the instance sizes a simulator reaches. The margins are large. On the entanglement and oracle problems the classical routine finishes in at most 12.3 microseconds, while the quantum framework takes 0.39 to 1.3 ms on the entanglement problems and 4.8 to 7.1 ms on the oracle problems, a gap of about 100 to more than 1000 times dominated by circuit construction and simulation overhead. On the variational problems the gap is starkest: classical MaxCut by brute force takes 0.02 ms against QAOA’s 266 ms, and the classical VQE and quantum-machine-learning baselines finish in 0.76 and 1.97 ms against 420 and 1274 ms for their quantum counterparts. Even the two algorithms whose entire fame is a speedup lose here: Shor’s order-finding factors 15 in 137 ms while trial division does it in 3 microseconds, and Grover’s search costs more wall-time than a classical scan of eight items. This is not a bug and not a slow implementation; it is the expected consequence of asking small-instance questions, where a quantum algorithm’s asymptotic advantage is nowhere near paying for the constant-factor overhead of building and simulating a circuit.

IV. WHERE QUANTUM WINS: PHYSICS, NOT WALL-TIME

That classical wins the wall-clock does not mean quantum does nothing; it means the win is in physics, not practical computation, and the benchmark shows exactly where (Fig. 2).

TABLE I
THE TWENTY PROBLEMS, REPRESENTATIVE INSTANCE: CLASSICAL VS BEST-QUANTUM WALL-TIME (MS) AND THE QUANTUM FRAMEWORK(S). CLASSICAL IS FASTER ON ALL TWENTY. FULL COSTS (QUBITS, GATES, SHOTS) ARE IN THE ARCHIVED TRACES.

Problem	category	classical (ms)	quantum (ms)
single-qubit	fundamentals	< 0.001	0.22
qrmg	fundamentals	< 0.001	0.22
interference	fundamentals	< 0.001	0.45
superdense	entanglement	< 0.001	0.39
teleportation	entanglement	< 0.001	1.26
chsh	entanglement	< 0.001	0.82
state-prep	entanglement	0.012	1.29
deutsch-jozsa	oracle	0.007	4.81
bernstein-vazirani	oracle	0.008	7.08
simon	oracle	0.012	4.94
grover	flagship	0.005	4.96
qft	flagship	1.42	4.91
qpe	flagship	0.068	6.66
shor	flagship	0.003	137.4
maxcut (QAOA)	variational	0.024	266.0
vqe	variational	0.76	420.6
qml	variational	1.97	1273.6
qec-repetition	noise/QEC	0.002	7.6
qec-surface	noise/QEC	0.002	7.5
noise	noise/QEC	0.45	13.3

The clearest case is the CHSH/Bell test [9]. With entangled qubits and the optimal measurement settings the correlator sum reaches $S = 2.83$, the Tsirelson bound [10], and *violates* the classical bound of 2 that any local-hidden-variable theory must obey, exactly the violation whose experimental realisation [11] earned the 2022 Nobel Prize. This is a quantum-beats-classical result, but it is a statement about the structure of nature (local realism is false), not a faster computation, and it requires

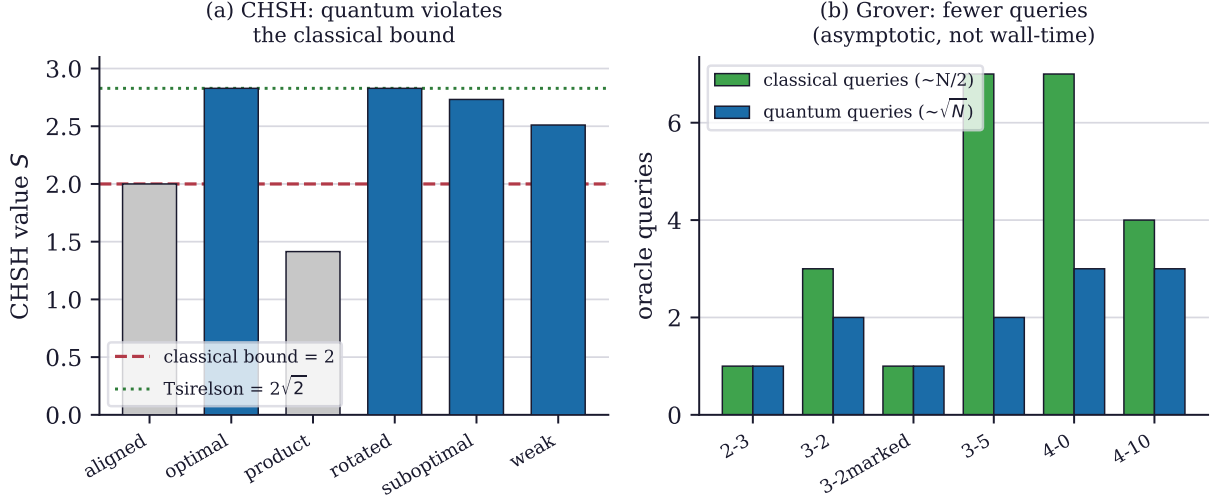


Fig. 2. The quantum phenomena the benchmark demonstrates: real, but not yet a wall-time advantage. **(a)** The CHSH value S across measurement settings: the optimal and rotated settings reach the Tsirelson bound $2\sqrt{2} \approx 2.83$ and *violate* the classical local-hidden-variable bound of 2, but only with entanglement; a product (separable) state gives $S = 1.41$ and the aligned setting sits exactly at 2. **(b)** Grover’s oracle-query count versus a classical scan: quantum uses provably fewer queries (two versus seven on the three-qubit, one-marked instance), the quadratic $\sqrt{N}$ advantage, which is real in query complexity even though it does not beat the classical wall-time at this N .

entanglement: the benchmark’s product-state variant gives $S = 1.41$ and does not violate anything. The second case is query complexity: Grover’s algorithm [4] finds a marked item in about $\sqrt{N}$ oracle queries against a classical scan’s $N/2$, and the benchmark measures the advantage directly (two quantum queries against seven classical on one instance). This too is real, a proven separation in the number of oracle calls, and it too fails to beat wall-time at reachable N because each quantum query costs far more to simulate than a classical array lookup. Shor’s algorithm [5] completes the picture: it factors 15 correctly through order-finding, a correct execution of a quantum algorithm, while remaining, at the scale that matters, no threat: factoring RSA-2048 would need on the order of a million noisy physical qubits with full fault tolerance [6], which is why Shor’s algorithm is not a near-term cryptographic danger.

V. DISCUSSION

The benchmark separates two questions. *Does quantum mechanics do things classical physics cannot?* Yes, and the CHSH violation is the clean demonstration. *Does a quantum computer beat a classical one in wall-clock time on these problems, at the instance sizes a simulator reaches?* No, on all twenty problems, and the wall-times say so, usually by two to four orders of magnitude. The two answers are not in tension, and putting them side by side shows a learner both the physics and the cost of computing with it at this scale. The report also makes a methodological point: a quantum-vs-classical comparison is only meaningful with the classical baseline actually run and its cost reported, and an asymptotic advantage cannot be observed on an instance far too small for it to appear. The benchmark builds this requirement into the architecture: every quantum solver has a classical sibling, and both costs are always on screen.

VI. SCOPE AND LIMITATIONS

Every trace runs on a *classical simulator* of the quantum circuit, not on quantum hardware; this is the regime a public laboratory can reach and is the correct regime for the lab-scale question, but it means the wall-times measure simulation cost, not a physical quantum processor, and a fault-tolerant machine at large scale would change the asymptotic story for Shor and for quantum chemistry (not demonstrated here, and not claimed). The instances are small (a handful of qubits), by necessity, so the benchmark measures the small- N regime where asymptotic advantages are absent by construction; it does not and cannot refute the existence of a large- N advantage, only show that it is not present at lab scale. The noise and error-correction cases are idealised-to-realistic simulator models, not device calibration data. And the frameworks are pinned versions; a newer release could shift the constants. The laboratory states the same scope on every panel; the report measures the small-scale regime and makes no claim about the ultimate limits of quantum computing.

VII. CONCLUSION

Run with the classical baseline beside every quantum method and both costs on the table, twenty canonical quantum problems on five real frameworks give one answer: at lab scale the classical baseline wins the wall-clock on all twenty, including the flagship speedup algorithms, because their advantages are asymptotic and the instances are small. And yet quantum wins at physics: the CHSH bound is violated, Grover’s query advantage is real, Shor factors correctly. On these problems, quantum methods beat classical ones at nonlocality and query complexity but not in wall-clock time at lab scale, and the benchmark shows this reproducibly, with measured numbers on real frameworks.

APPENDIX A REPRODUCTION

The benchmark is the 119 traces archived under `data/artifacts/`, one folder per problem, each trace recording the problem, the solvers, the per-solver costs, and the quantum-vs-classical outcome. The aggregation script (`manuscripts/quantum-vs-classical/figures/aggregate.py`) walks them into the summary artifact the figures read, extracting the per-problem wall-times, the CHSH value across settings, and the Grover query counts. Re-running the traces (each is seeded) reproduces the numbers; the figures and Table I regenerate from the archived summary (`manuscripts/quantum-vs-classical/data/bench_summary.json`).

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